

AN ADVENTURE BECKONS
Shubhanshu Shukla (2nd from left) and fellow crew members aboard the SpaceX Dragon spacecraft before lift-off, Jun. 25

powerful, long-lasting rover to explore resources for human sustenance on the lunar surface.

In doing so, India will join the global Moon Rush, with the US sanctioning the Artemis projects, which will mark American astronauts' return to the moon after a hiatus of over 50 years. China has been actively exploring the Moon, even robotically collecting lunar samples and flying them back for analysis. Half a dozen other countries, including Russia, Japan and South Korea, have joined the lunar bandwagon with ambitious flights. This coincides with the increasing interest within countries to establish space stations of their own. Russia has announced its withdrawal from the ISS and plans to develop its own Russian Orbital Service Station (ROSS). Apart from India and China, there are three major private players who are keen to develop commercial space stations. These include Orbital Reef, a joint project by Blue Origin and Sierra Space; Starlab by Voyager Space and Airbus; and Axiom. Jitendra Singh, the Union minister of state for science and technology, earth sciences and space, says, "From a follower, India intends to be a frontliner in space. India is now an equal global partner in space endeavours."

To trigger an unprecedented boost in private participation in Indian space, the Modi government in June 2020, in the thick of the Covid pandemic, ended ISRO's monopoly to build and launch rockets and satellites. It not only allowed entry to private players but also opened up ISRO facilities for them to use on payment. Subsequently, it also allowed 100 per cent FDI in the space sector. Close to 200 space start-ups have come up in the past four years, and many of them have been extremely successful. This is in keeping with the international trend of private giants like Elon Musk's Space-X and Jeff Bezos's Blue Origin, competing with established giants like Boeing and Lockheed Martin.

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India's space market was worth $8.4 billion (Rs 71,800 crore) in 2024, just 2 per cent of the global space market, way below its potential. The Modi government is pushing for an aggressive target of reaching $44 billion (Rs 3.75 lakh crore) by 2033, accounting for 8 per cent of the global share, with its ambitious human space exploration mission being a major driver. Up in space, Shukla's advice to youngsters is, "There is no one path to success, but there is one common thing—Never Stop Trying." That is something India, too, can follow in its giant ambitions for human spaceflight in the wake of Shukla's sojourn. ■